MATH 152 **QUIZ 6A** NAME Solutions

Mrs. Bonny Tighe 25 points

7.8,8.1,8.2 TA_____ Wed. 10/17/2018

1. State the Comparison Test for Improper Integrals

2pts Suppose f + g are continuous functions with $f(x) \geq g(x) \geq 0$ for $x \geq a$. i) If $\int_a^\infty f(x)\,dx$ is convergent then $\int_a^\infty g(x)\,dx$ is convergent

ii) If $\int_a^\infty g(x)\,dx$ is divergent then $\int_a^\infty f(x)\,dx$ is divergent

1pt 2. The area under a curve can be found if the improper definite integral is Convergent

3. Describe the two types of improper integrals.

1pt 1- a limit of integration is $\pm\infty$

2- the function is not defined on the interval

4. Determine whether each integral is convergent or divergent. Evaluate those that are convergent.

3pts a) $\int_0^\infty x e^{-x}\,dx$

$u = x \quad dv = e^{-x}dx$

$du = dx \quad v = -e^{-x}$

$\lim_{t\to\infty} \int_0^t x e^{-x}\,dx$

$\lim_{t\to\infty} \left[-xe^{-x} - \int -e^{-x}\,dx\right]$

$\lim_{t\to\infty} \left[-\frac{x}{e^x} - \frac{1}{e^x}\right]_0^t$

$\lim_{t\to\infty} \left[\left(\frac{-t}{e^t} - \frac{1}{e^t}\right) - \left(\frac{0}{e^0} - \frac{1}{e^0}\right)\right]$

$0 - 0 - 0 + 1 = 1$

so convergent

b) $\int_0^1 \frac{1}{\sqrt{1-x^2}}\,dx$ 3pts

$\lim_{t\to 1^-} \int_0^t \frac{1}{\sqrt{1-x^2}}\,dx$

$\lim_{t\to 1^-} \sin^{-1}x \Big]_0^t$

$\lim_{t\to 1^-} (\sin^{-1}t - \sin^{-1}0)$

$\sin^{-1}1 = \pi/2$ $= \pi/2$

so convergent

5. Use the comparison test to determine whether the integrals diverge or converge.

(3pts) $\int_1^\infty \frac{x}{\sqrt{x^5+4x+3}}dx$

Compare to $\int_a^\infty \frac{1}{x^{3/2}}dx$ which is convergent $p = 3/2 > 1$

$\frac{1}{x^{3/2}} \geq \frac{x}{\sqrt{x^5+4x+3}}$

$\sqrt{x^5+4x+3} \geq x^{5/2}$

$x^5+4x+3 \geq x^5$

$4x+3 \geq 0$ which is true for $x \geq 1$

So $\int_1^\infty \frac{x}{\sqrt{x^5+4x+3}}dx$ is also convergent

(4pts) 6. Find the length of the arc of the curve $x = \ln(y^2-1)$, $2 \leq y \leq 4$

$L = \int_2^4 \sqrt{1+(dx/dy)^2}\,dy$ $\quad dx/dy = \frac{1}{y^2-1}(2y)$

$L = \int_2^4 \sqrt{1+\frac{4y^2}{(y^2-1)^2}}\,dy = \int_2^4 \sqrt{\frac{(y^2-1)^2+4y^2}{(y^2-1)^2}}\,dy$

$= \int_2^4 \frac{\sqrt{y^4-2y^2+1+4y^2}}{y^2-1}dy$ ← one step $= \int_2^4 \frac{\sqrt{y^4+2y^2+1}}{y^2-1}dy$ $\quad \sqrt{(y^2+1)^2}$

$\int \frac{y^2+1}{y^2-1}dy$ $\quad y^2-1 \overline{)\,y^2+1}$ $\quad \frac{1}{-(y^2-1)}$ $\quad 2$

$\int_2^4 1+\frac{2}{(y+1)(y-1)}dy$

$\frac{A}{y+1}+\frac{B}{y-1} = \frac{2}{(y+1)(y-1)}$

$A(y-1)+B(y+1)=2$

$y=1$ $\quad B=1$

$y=-1$ $\quad A=-1$

$\int_2^4 \left(1+\frac{1}{y-1}-\frac{1}{y+1}\right)dy$

$y+\ln\left|\frac{y-1}{y+1}\right|\Big]_2^4 = \left(4+\ln\left|\frac{3}{5}\right|\right)-\left(2+\ln\left|\frac{1}{3}\right|\right)$

$2+\ln\left|\frac{3/5}{1/3}\right| = 2\ln\left|\frac{9}{5}\right|$

2pts

7. Set up an integral, but do not solve, for the area of the surface obtained by rotating the curve

a) $y = \ln x$, $1 \le y \le 3$ about a) the x-axis and b) the y-axis

$x = e^y$ dy $r = y$ $r = x$

a) $2\pi \int_1^3 y\sqrt{1 + e^{2y}}\, dy$ $dx/dy = e^y$

b) $2\pi \int_1^3 e^y \sqrt{1 + e^{2y}}\, dy$

2pts

b) $y = \sin x$, $0 \le x \le \pi$ about a) the x-axis and b) the y-axis

dx $r = y$ $r = x$

a) $2\pi \int \sin x \sqrt{1 + (\cos x)^2}\, dx$ $dy/dx = \cos x$

b) $2\pi \int x \sqrt{1 + \cos^2 x}\, dx$

4 pts

8. The given curve is rotated about the x-axis. Find the area of the resulting surface.

$r = y$

$x = \frac{y^3}{3} + \frac{1}{4y}, \quad 1 \le y \le 2 \qquad S = 2\pi \int_a^b radius\ arclength$

$x = \frac{1}{3}y^3 + \frac{1}{4}y^{-1}$

$dx/dy = \frac{3}{3}y^2 + -\frac{1}{4}y^{-2}$

$= y^2 - \frac{1}{4y^2}$

$(dx/dy)^2 = y^4 - 2(y^2)(\frac{1}{4y^2}) + \frac{1}{16y^4}$

$y^4 - \frac{1}{2} + \frac{1}{16y^4}$

$S = 2\pi \int_1^2 y \sqrt{1 + (dx/dy)^2}\, dy$

$= 2\pi \int_1^2 y \sqrt{1 + y^4 - \frac{1}{2} + \frac{1}{16}y^4}\, dy$

one fourth

$2\pi \int_1^2 y \sqrt{y^4 + \frac{1}{2} + \frac{1}{16y^4}}\, dy$

$2\pi \int_1^2 y \sqrt{(y^2 + \frac{1}{4y^2})^2}\, dy$

$2\pi \int_1^2 y(y^2 + \frac{1}{4y^2})\, dy = 2\pi \int_1^2 y^3 + \frac{1}{4} \cdot \frac{1}{y}\, dy$

$2\pi \left[\frac{1}{4}y^4 + \frac{1}{4}\ln|y|\right]_1^2 = 2\pi \left[\frac{1}{4}(16) + \frac{1}{4}\ln(2) - (\frac{1}{4} + \frac{1}{4}\ln(1))\right]$

$2\pi \left[4 + \frac{1}{4}\ln|3| - \frac{1}{4}\right]$

MATH 152
Mrs. Bonny Tighe

QUIZ 6
25 points
7.8,8.1,8.2

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1. The area under a curve can be found if the improper definite integral is Convergent

2. Describe the two types of improper integrals.

1- a limit of integration is $\pm\infty$

2- the function is not defined at some point on the interval

3. State the Comparison Test for Improper Integrals Suppose f & g are continuous functions and $f(x) \geq g(x) \geq 0$ for $x \geq 0$ i) If $\int_a^b f(x)dx$ is convergent then $\int_a^\infty g(x)dx$ is convergent ii) If $\int_a^b g(x)dx$ is divergent then $\int_a^\infty f(x)\,dx$ is divergent

4. Determine whether each integral is convergent or divergent. Evaluate those that are convergent.

a) $\int_1^\infty \frac{1}{x^2+1}\,dx$

$\lim_{t\to\infty} \int_1^t \frac{1}{x^2+1}dx =$

$\lim_{t\to\infty} \left[\tan^{-1}x\right]_1^t$

$\lim_{t\to\infty} \left[\tan^{-1}t - \tan^{-1}1\right]$

$\pi/2 - \pi/4$

$\pi/4$ So Converges

b) $\int_1^3 \frac{1}{x^2-2x-3}dx = \int_1^3 \frac{1}{(x-3)(x+1)}dx$

$\frac{A}{x-3} + \frac{B}{x+1} = \frac{1}{(x-3)(x+1)}$

$A(x+1) + B(x-3) = 1$

$x=-1$ $B = -1/4$

$x=3$ $A = 1/4$

$\lim_{t\to 3^-} \int_1^t \left(\frac{1/4}{x-3} - \frac{1/4}{x+1}\right)dx =$

$\lim_{t\to 3^-} \frac{1}{4}\ln\left|\frac{x-3}{x+1}\right|\Big]_1^t$

$\lim_{t\to 3^-} \frac{1}{4}\ln\left|\frac{t-3}{t+1}\right| - \frac{1}{4}\ln\left|\frac{-2}{2}\right|$

$\frac{1}{4}\ln|0| \leftarrow$ DNE

So Divergent

5. Use the comparison test to determine whether the integrals diverge or converge.

$$\int_3^\infty \frac{x}{\sqrt{x^3-4x-3}}\,dx$$

Compare to $\int_3^\infty \frac{1}{x^{1/2}}\,dx$ which is divergent $p = 1/2 \le 1$

$$\frac{x}{\sqrt{x^3-4x-3}} \ge \frac{1}{x^{1/2}}$$

$x^{3/2} \ge \sqrt{x^3-4x-3}$

$x^3 \ge x^3-4x-3$

$4x+3 \ge 0$ which is true for $x \ge 3$

So $\int_3^\infty \frac{x}{\sqrt{x^3-4x-3}}\,dx$ is also divergent

6. Find the length of the arc of the parabola $y = \frac{x^2}{4} - \frac{1}{2}\ln(x)$, $1 \le x \le 2$

$y = \frac{1}{4}x^2 - \frac{1}{2}\ln x$

$dy/dx = \frac{2}{4}x - \frac{1}{2x}$

$(dy/dx)^2 = \left(\frac{1}{2}x - \frac{1}{2x}\right)^2$

$\frac{1}{4}x^2 - 2\left(\frac{1}{2}x\left(\frac{1}{2x}\right)\right) + \frac{1}{4x^2}$

$\frac{1}{4}x^2 - \frac{1}{2} + \frac{1}{4x^2}$ one of these

$L = \int_1^2 \sqrt{1+(dy/dx)^2}\,dx$

$L = \int_1^2 \sqrt{1+\frac{1}{4}x^2-\frac{1}{2}+\frac{1}{4x^2}}\,dx$

$L = \int_1^2 \sqrt{\frac{1}{4}x^2+\frac{1}{2}+\frac{1}{4x^2}}\,dx = \int_1^2 \sqrt{\left(\frac{1}{2}x+\frac{1}{2x}\right)^2}\,dx$

$\int_1^2 \left(\frac{1}{2}x + \frac{1}{2}\cdot\frac{1}{x}\right)dx = \frac{1}{4}x^2 + \frac{1}{2}\ln|x| \Big]_1^2$

$\left(\frac{1}{4}(4) + \frac{1}{2}\ln|2|\right) - \left(\frac{1}{4} + \frac{1}{2}\ln|1|\right)$

$1 + \frac{1}{2}\ln|2| - \frac{1}{4} + 0 = \frac{3}{4} + \frac{1}{2}\ln|2|$

$r = x$

7. The given curve is rotated about the y-axis. Find the area of the resulting surface.

$x = \sqrt{e^y + 1} \quad 1 \le y \le 4 \qquad S = 2\pi \int_a^b radius\ arclength$

dy

$x = (e^y + 1)^{1/2}$

$dx/dy = \frac{1}{2}(e^y+1)^{-1/2}(e^y)$

$= \frac{e^y}{2\sqrt{e^y+1}}$

$(dx/dy)^2 = \frac{e^{2y}}{4(e^y+1)}$

$= 2\pi \int_1^4 \sqrt{e^y+1}\sqrt{1 + \frac{e^{2y}}{4(e^y+1)}}\, dy$

$S = 2\pi \int_1^4 \sqrt{e^y+1}\sqrt{\frac{4e^y+4+e^{2y}}{4(e^y+1)}}$

$S = 2\pi \int_1^4 \sqrt{e^y+1}\,\frac{\sqrt{(e^y+2)^2}}{2\sqrt{e^y+1}}\, dy$

$S = \pi \int \sqrt{(e^y+2)^2}\, dy = \pi \int (e^y+2)\, dy$

$\pi \left[e^y + 2y\right]_1^4 = \pi\left[(e^4+8) - (e^1+2)\right]$

$\pi\left[e^4 + 6 - e\right]$

8. Set up an integral, but do not solve, for the area of the surface obtained by rotating the curve

a) $y = \cos x, \quad 0 \le x \le \pi/2$ about a) the x-axis and b) the y-axis

$r = y$ $\quad$ $r = x$

a) $2\pi \int_0^{\pi/2} \cos x \sqrt{1 + (-\sin x)^2}\, dx$

b) $2\pi \int_0^{\pi/2} x \sqrt{1 + \sin^2 x}\, dx$

b) $y = \ln x, \quad 1 \le y \le 3$ about a) the x-axis and b) the y-axis

$r = y$ $\quad$ $r = x$

$y = \ln x$

$e^y = x$

a) $2\pi \int_1^3 y \sqrt{1 + e^{2y}}\, dy$

b) $2\pi \int_1^3 e^y \sqrt{1 + e^{2y}}\, dy$